

verters and transceivers can shorten development time and therefore has a ripple-through effect of lowering the chip's total development cost. However, even though design time is reduced, there can be other tradeoffs that must be considered. Standard analogue library cells do not pack as densely as digital cells. Using analogue library cells can result in blocks of unused silicon on the die that will needlessly lower the number of potential dies on a wafer.

Additionally, since analogue circuitry of a mixed-signal ASIC is likely to be the input or output of the circuit, or both, these cells must be oriented closer to the periphery of the chip to facilitate easy access to bonding pads.

Handcrafting some or all analogue functions allows you to accomplish several things. In a mixed-signal design, handcrafted analogue circuits are laid out to fill voids created when using standard cells, better optimising overall silicon area utilisation.

Moreover, handcrafting the analogue portion allows you to determine precisely the performance parameters of the circuit rather than be restricted to the fixed performances of a limited number of standard cells available in the library.

Myth #5. Cell based ASIC designs ensure product differentiation.

FALSE. Designing the analogue portion of a mixed-signal ASIC using a cell library is tantamount to designing a system using standard, off-the-shelf, analogue ICs, with one key exception—limited selection. At the board level, there are tens of thousands of IC amplifiers, voltage references, converters and more to choose from. In a cell library, you are limited to choosing from a few dozen amplifiers, voltage references, converters, etc. Performance com-

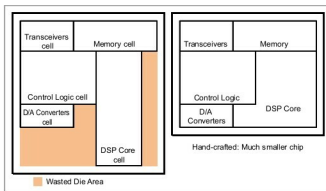


Fig. 3: Benefits of handcrafted analogue

promises may be needed to accommodate these limited choices.

Analogue-centric ASIC development affords a perfect opportunity to rise above your competition. As noted earlier, nearly 60 per cent of the worldwide analogue IC market is for ASICs. If you and your competitors are basing your designs around the same mixed-signal cell libraries, both of you will have approximately the same performance specifications, dictated by the specifications of the library cells.

True product differentiation comes from invention. It is derived by creating uniqueness to a product not readily available to the competition. Cell libraries fail to deliver the necessary uniqueness often needed in critical analogue applications.

Myth #6. Handcrafted analogue is too expensive, compared to standard cells.

FALSE. NRE costs are a compilation of several variables. These costs must be amortised over the number of chips produced during the lifetime of the product to determine their effect on the unit cost of the ASIC. When executed properly, NRE costs associated with handcrafting the analogue circuitry return a disproportionately lower unit cost of the final chip. The key to success is analogue design experience resident at the ASIC semiconductor company doing the integration.

Handcrafted analogue can

create the differentiation required to break out of the pack with a superior-performing chip and, thus, a superior end product for the customer. Additionally, stepping back from the cell library approach opens up options for manufacturing, since cell libraries are typically developed for one process at one fab.

Broad-use libraries are available that specify a process, for example, 0.35µm CMOS, but have relaxed specifications such that these can be instantiated in multiple fabs.

Handcrafted analogue creates an unlimited set of manufacturing options, especially through the use of boutique foundries. Many of the boutique fabs differentiate themselves by the variety of services they offer and their willingness to make adjustments to their processes to accommodate optimisation of the chip's performance.

Conclusion

The application will always determine the appropriate combinations of technologies that are best suited for the ASIC design. As our dependence on cognitive prosthesis devices (smartphones, Wii controllers, tablet PCs, the IoT, etc) increases, copper tethers disappear and analogue increases its dominance in ASIC designs. MEMS advances have placed Star Trek style sensors in our daily lives. Medical imaging, sensing and monitoring continues to improve our daily lives. All of these more increasingly rely upon better, faster, denser analogue circuits.

When considering a new ASIC design, carefully consider the role analogue will play in its deployment. To minimise risk, seek out an analogue ASIC partner with the right analogue design skills and experience to match the application. **EFY**